Neural Signals and Signal Processing – Miniproject 1

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1 Part 1 – Practicals

1.1 Structural preprocessing

a) Skull-stripping. Non-brain tissue is removed because it would bias segmentation and registration. We used FSL BET (Smith, 2002) and compared several settings visually (fractional threshold $f \in \{0.5, 0.45, 0.4\}$, robust centre `-R`, vertical gradient g). We kept $f = 0.5$ with `-R`: it gave the best cortical coverage, whereas $f = 0.4$ kept non-brain tissue and $g > 0$ cut the superior cortex (Fig. 9, Appendix). Small residual errors (interhemispheric fissure, posterior region) were left uncorrected given the 2 mm functional resolution (Fig. 1a).

b) Segmentation. Splitting the brain into CSF, grey matter (GM) and white matter (WM) allows tissue-specific analyses. FSL FAST (Zhang et al., 2001) with default settings follows the cortical folding well (Fig. 1b), but deep GM (thalamus, basal ganglia) is partly labelled as WM, which is a known limitation of intensity-based methods.

c) Registration. Registration to MNI152 (1 mm) puts the subject in a common space for atlas comparison. FLIRT (12 DOF, normalized correlation, the best cost function tested (Fig. **??**) gave $r = 0.63$ with the template; since an affine transform cannot absorb inter-individual differences, ANTs SyN (Avants et al., 2008) raised it to $r = 0.86$ (Fig. 1c). The Jacobian determinant stayed positive in the brain, so no folding occurred (Fig. 11a). The same transform was applied to the segmentation.

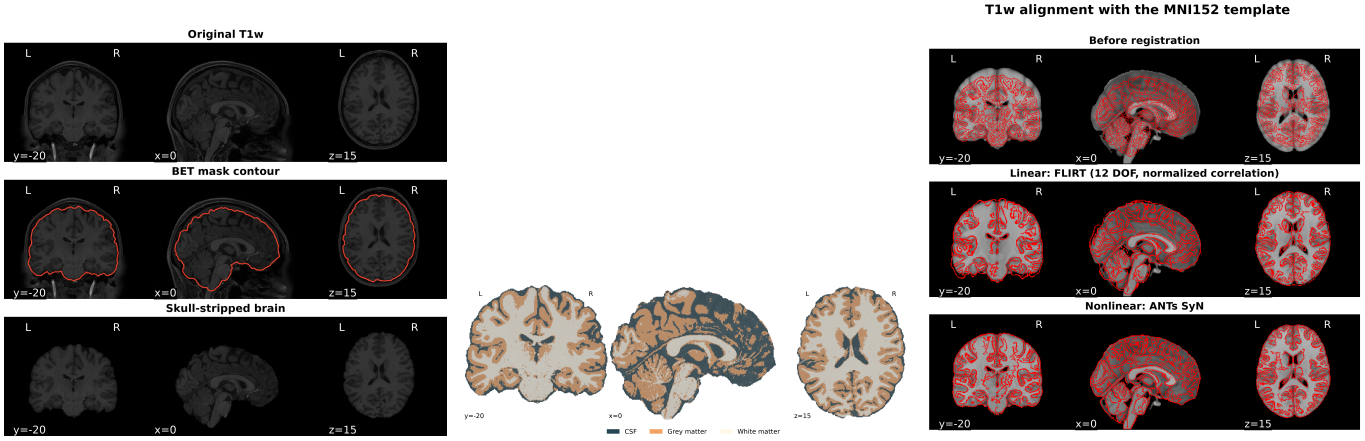


Figure 1: Structural preprocessing. (a) T1w, BET mask and skull-stripped brain; (b) FAST segmentation; (c) alignment with MNI152 before, after FLIRT and after ANTs SyN.

1.2 Functional preprocessing

a) Motion correction. Head motion creates spurious signal changes, so volumes were realigned to the middle volume with MCFLIRT (6 rigid-body parameters), and Power framewise displacement (FD, 50 mm radius) was computed. Motion is very small (mean FD 0.085 mm, max 0.259 mm). The threshold $Q_3 + 1.5\,\mathrm{IQR}$ (0.233 mm) flags only three volumes (25, 26, 45), early in the run (Fig. 2a). We kept all volumes and saved the motion and spike regressors for the GLM.

b) Smoothing. A Gaussian kernel of 6 mm FWHM ($\sigma \approx 2.55$ mm) increases the signal-to-noise ratio and the sensitivity to extended activations, at the cost of spatial detail; Fig. 2b (volume 158) shows less local variation and softer tissue boundaries.

c) Anatomical–functional alignment. The unsmoothed fMRI overlaid on the MNI T1w (Fig. 2c) shows well-matching outer boundaries and no large displacement, with only small discrepancies in inferior regions. Functional-to-anatomical registration and MNI normalisation can therefore be skipped, as the data are already in MNI space (assessment limited to one volume and axial slices).

1.3 GLM

a) Design matrix. The GLM models each voxel's time course as a weighted sum of regressors. As the dataset contains a single subject and run, it was fitted at the first level. Only the block-level conditions (story, math) were kept, and their nested sub-steps (presentation, question, response) and cues were removed;

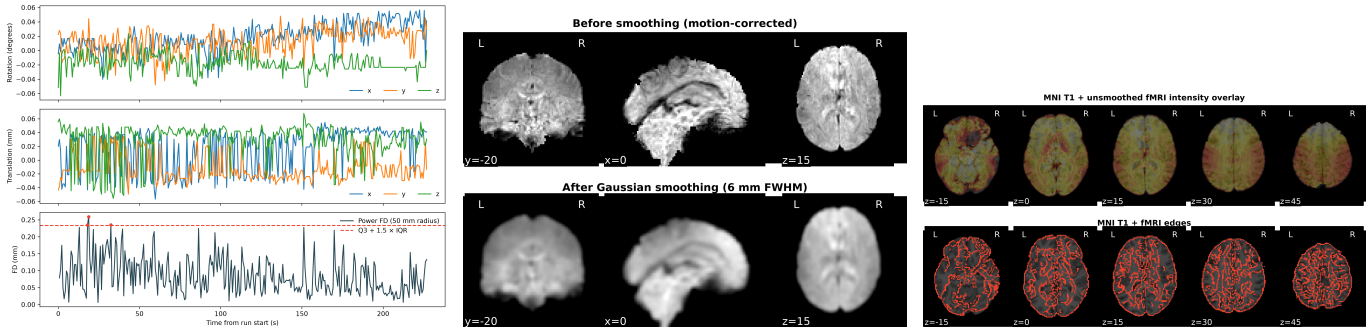


Figure 2: Functional preprocessing. (a) Motion parameters and FD (dashed: threshold); (b) volume 158 before and after smoothing; (c) fMRI overlay on the MNI T1w.

both regressors were convolved with the canonical SPM HRF to account for the hemodynamic delay. The nuisance regressors were taken from the functional preprocessing (Section 1.2a). Slow scanner drifts were modelled with a cosine basis set. The high-pass cut-off was set to 125 s rather than 100 s, as the latter produced a drift term correlated with the task (see below). An AR(1) model accounted for the temporal autocorrelation of the noise. Story and math blocks alternate with only brief pauses between them, leaving almost no rest period (Fig. 12). (Fig. 3a)

b) Regressor correlation. Story and math are strongly anticorrelated ($r = -0.94$) (Fig. 3b), because the two conditions alternate with almost no rest. Task regressors are only weakly correlated with motion ($|r| \leq 0.18$) and spike regressors ($|r| \leq 0.06$), so there is no task-related motion. With a 100 s high-pass cut-off, the fastest drift term correlated with the task ($|r| = 0.42$) because of the irregular block lengths. Lowering the cut-off to 125 s removed it ($|r| \leq 0.22$). Correlations among motion parameters and with drifts are expected and do not affect the task estimates.

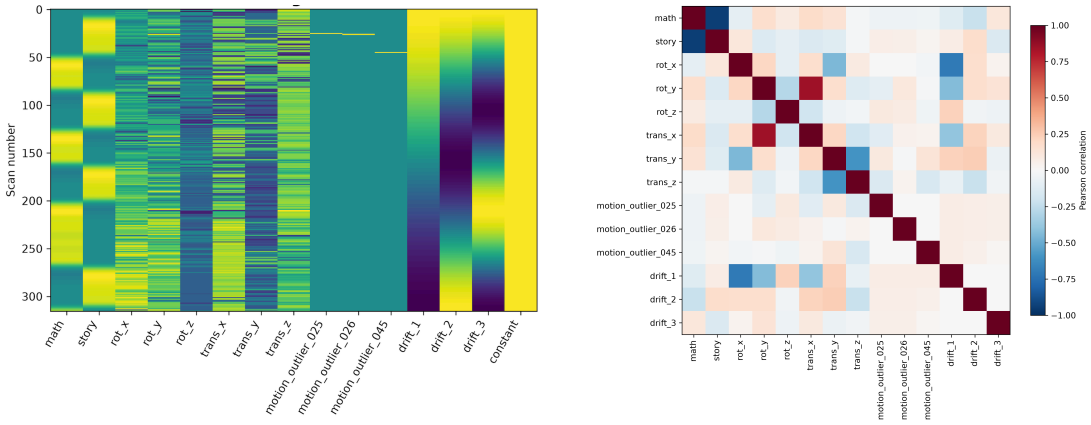


Figure 3: GLM design. (a) Design matrix: story and math regressors, six motion parameters, three spike regressors, cosine drifts and constant. (b) Correlation between design-matrix regressors (constant excluded).

c) Single-condition maps. As stated above, the GLM was run at the first (single-subject) level. Each task map tests whether the condition's signal differs from the implicit baseline, in this case the periods not covered by any task regressor (only brief cues and pauses between blocks). These z-maps (FDR $q < 0.05$, clusters > 10 voxels; Fig. 4) show mainly widespread negative effects, overlap substantially in superior slices (left parietal and medial frontal cortex), and lack any positive response in auditory cortex despite the auditory nature of both tasks. These maps are thus not interpretable as task activity: they result from the design, which contains almost no rest period and hence no reliable baseline. Meaningful inference therefore relies on the story-math contrast, in which the baseline cancels out (see below).

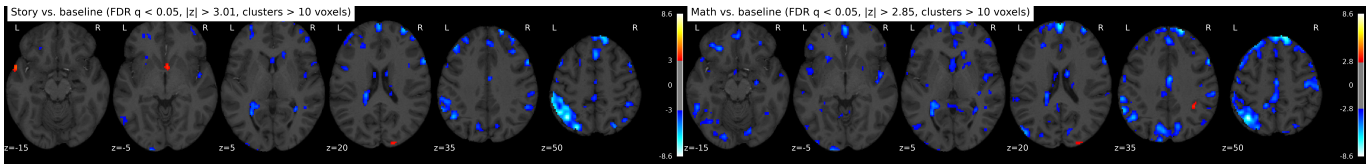


Figure 4: Z-maps of the story (top) and math (bottom) regressors against the implicit baseline (FDR $q < 0.05$, clusters > 10 voxels).

d) Story vs. math contrast. The contrast vector $c = [-1, +1, 0, \ldots, 0]$ (math -1, story $+1$, all nuisance regressors and the constant 0; Fig. 5) compares the two conditions directly, so that the poorly implicit baseline cancels out. Unlike the single-condition maps, the contrast map (FDR $q < 0.05$, $|z| > 2.33$, clusters > 10 voxels) shows a clear dissociation: story > math effects (red) dominate medial frontal, posterior medial, temporal and occipital cortex, whereas math > story effects (blue) are concentrated in lateral parietal

and lateral prefrontal cortex. Effects are spatially extensive, as the two tasks differ strongly and the FDR threshold, which adapts to the data, is relatively low ($|z| > 2.33$).

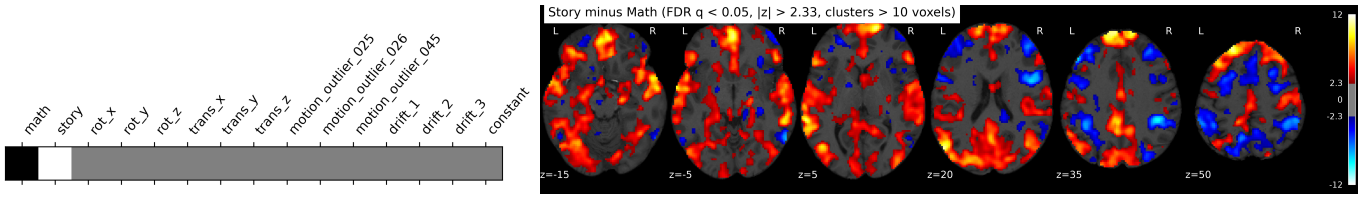


Figure 5: Story vs. math contrast. (a) Contrast vector: -1 on math, $+1$ on story, 0 on all other regressors. (b) Resulting z-map (FDR $q < 0.05$, $|z| > 2.33$, clusters > 10 voxels); red: story > math, blue: math > story.

e) Functional networks. To interpret the contrast in terms of resting-state networks, the thresholded map was compared voxel-wise with the Yeo et al. (2011) 7-network atlas, and the percentage of significant voxels was computed for each network (Fig. 6). Semantic processing (story > math) mainly engaged the default mode (57.6% of its voxels) and limbic (29.2%) networks. The default-mode effect may partly reflect deactivation during math rather than activation during story. Together with the medial frontal, posterior medial and temporal effects, the story > math pattern overlaps with parts of the known semantic network (Binder et al., 2009). Numerical processing (math > story) mainly engaged the dorsal attention (29.8%) and frontoparietal (29.2%) networks, which include the intraparietal sulcus, a core region for number processing (Dehaene et al., 2003). The somatomotor network (45.9% story > math) includes auditory cortex in this atlas (Yeo et al., 2011), likely reflecting the longer speech input of the stories compared to the short math problems (Binder et al., 2011). The visual network effect (51.2%) was unexpected, as neither task is visual. This could reflects visual imagery during the stories. The ventral attention network showed mixed effects (19.8% story > math, 14.9% math > story), consistent with its role in attention, by detecting relevant stimuli (Corbetta & Shulman, 2002), such as the questions present in both tasks. These percentages depend on the (relatively low) FDR threshold and ignore subcortical and cerebellar effects, as the atlas covers only the cortex. They should therefore be compared across networks rather than interpreted in absolute terms.

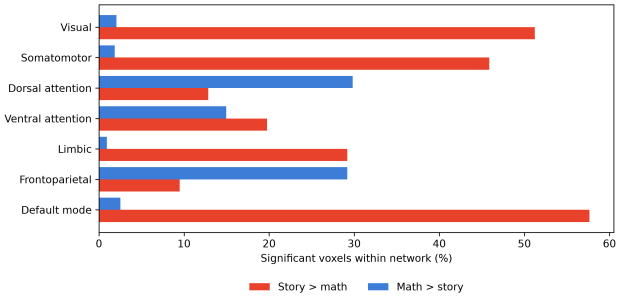


Figure 6: Percentage of voxels within each Yeo (2011) network showing significant story > math (red) or math > story (blue) effects (FDR $q < 0.05$).

f) Semantic regions. The semantic network is typically left-lateralized and includes the inferior parietal lobe, lateral, ventral and anterior temporal cortex, inferior frontal gyrus (IFG), medial prefrontal cortex and posterior cingulate cortex (Binder et al., 2009). Our story > math map recovers the medial prefrontal, posterior cingulate and lateral temporal components, but the effects are largely bilateral rather than left-lateralized, and the left IFG shows only partial story > math effects. Several factors may explain these differences. First, the math condition is itself auditory and verbal: phonological processing and verbal working memory, which engage the left IFG (Poldrack et al., 1999), are required by both tasks and thus cancel out in the contrast. Second, fMRI is less sensitive in parts of the semantic network, notably the ventral anterior temporal lobe, owing to susceptibility-related signal loss (possibly visible as a dark region in the right anterior part of the $z = -15$ slice, Fig. 2c), and whole-brain rather than region-of-interest analyses further reduce sensitivity (Visser et al., 2010). Finally, the bilateral pattern likely reflects single-subject variability, the 6 mm smoothing and the relatively low FDR threshold, which spread moderate effects into the right hemisphere.

1.4 Part 1 - Theoretical questions

Q1a. Areas stimulated by either story or math. This is the union of both effects against baseline. It can be tested with an F-test on both task regressors ($H_0 : \beta_{\text{math}} = \beta_{\text{story}} = 0$, contrast matrix with rows $[1, 0, 0, \dots]$ and $[0, 1, 0, \dots]$), which detects voxels where at least one task explains variance, whatever the sign. Alternatively, two one-sided t-tests ($[1, 0, \dots]$ and $[0, 1, \dots]$) can be combined, keeping voxels significant in at least one map (correcting for the two tests). This also shows which task drives each voxel. The map should include both shared regions (auditory, motor, attention) and task-specific ones (language

areas for story, parietal areas for math). In this dataset, however, both conditions cover almost the entire acquisition without rest, so the implicit baseline is poorly estimated and these maps are unreliable (cf. 3c).

Q1b. Question versus answer. The block regressors are replaced by one regressor per phase (presentation, question, response) for each task, to limit confounding between phases. The contrast $c = [1, -1, 1, -1]$ on $[\beta_{\text{story Q}}, \beta_{\text{story A}}, \beta_{\text{math Q}}, \beta_{\text{math A}}]$ (zeros elsewhere) gives positive values in areas driven by the auditory question (auditory/language cortex) and negative values in areas driven by the response (motor and premotor cortex, button press). Since it is a differential contrast, it does not depend on the poorly implicit baseline. Task-specific effects can be tested with $[1, -1, 0, 0]$ and $[0, 0, 1, -1]$, and their difference with $[1, -1, -1, 1]$ (interaction). As the answer closely follows the question, their hemodynamic responses overlap and the two regressors are partly collinear, which makes the estimates less precise.

Q1c. Processing versus preparation. The cue preceding each block marks task preparation, while the block itself (presentation, question, response) corresponds to processing. We add a cue regressor to the story and math block regressors and use $c = [-1, \frac{1}{2}, \frac{1}{2}]$ on $[\beta_{\text{cue}}, \beta_{\text{story}}, \beta_{\text{math}}]$, so that the weights sum to zero and the contrast does not depend on the implicit baseline. Positive voxels are more engaged in processing (auditory, language, arithmetic and motor areas) and the reverse contrast would be expected to show regions involved in task preparation, such as attentional control areas. However, the cue is very short (< 1 s) and immediately followed by the block, so its regressor is weak and partly collinear with the block regressors, which limits the precision of this contrast.

Q2. Importance of the story regressor. We use an F-test comparing the full model with a reduced model without the story regressor ($H_0 : \beta_{\text{story}} = 0$, i.e. it explains no extra variance). For a single regressor, this F-test is equivalent to a two-sided t-test on β_{story} ($F = t^2$). In the present data, however, story and math together cover almost the whole run, so story is nearly equal to the constant minus math (story–math correlation $r = -0.94$, Fig. **??**). The reduced model can therefore largely reproduce it, and the F-test only measures what story explains beyond math and the constant, which may be small even in regions strongly engaged by the task.

Q3. Adding a second run. We test the same hypothesis as in Section 3d, $H_0 : \beta_{\text{story}} - \beta_{\text{math}} = 0$ against $H_1 : \beta_{\text{story}} - \beta_{\text{math}} \neq 0$, assuming this effect is identical in both runs. Option 1: concatenate the runs in one GLM with common task regressors and run-specific constants, drifts and motion regressors, and test the contrast with a t-test on all volumes.Option 2: fit one GLM per run and combine the two contrast estimates by an inverse-variance weighted average. In both cases, increasing the amount of data reduces the variance of the estimate and thus increases statistical power. A second-level t-test across the two runs would not be appropriate, as two values give only one degree of freedom.

2 Part 2 – PCA (variant 1)

2.1 Practicals

1) PCA. PCA was applied to the preprocessed and smoothed fMRI run. The 4D data were reshaped into a matrix with 316 timepoints as samples and brain-mask voxels as features. Each voxel's temporal mean was subtracted before extracting the principal components. This allowed us to identify the dominant spatial patterns of variability and their associated temporal scores.

2) PCA Components. The explained variance ratio was examined using a scree plot to select the number of components (Fig. 7). The first five PCs were retained for visualization, accounting for 31.98% of the total variance. The variance decreased rapidly for the first components and more gradually afterward, supporting this choice. Their spatial loadings were reconstructed into 3D brain maps and displayed in axial view. PC1 showed widespread strong loadings, while PC3 exhibited distinct frontal and posterior clusters, suggesting potentially meaningful spatial patterns.

3) PCA–GLM correlation. Pearson correlations were computed between the temporal scores of the first 10 PCs and the GLM regressors, including task conditions, motion parameters, motion outliers, and temporal drifts. The resulting correlation matrix (Fig. 8) showed that PC1 was strongly associated with drift ($r = 0.79$) and rotation ($r = -0.58$), consistent with nuisance-related variability. PC3 showed the strongest task associations (math: $r = -0.70$; story: $r = 0.70$), and its map partly resembles the story – math contrast (Fig. 5). However, its correlation with motion ($r = -0.51$) suggests possible confounding. PC2 and PC4 also correlated with motion, whereas PC5 showed weaker associations. Overall, the components explaining the most variance were not necessarily the most task-related.

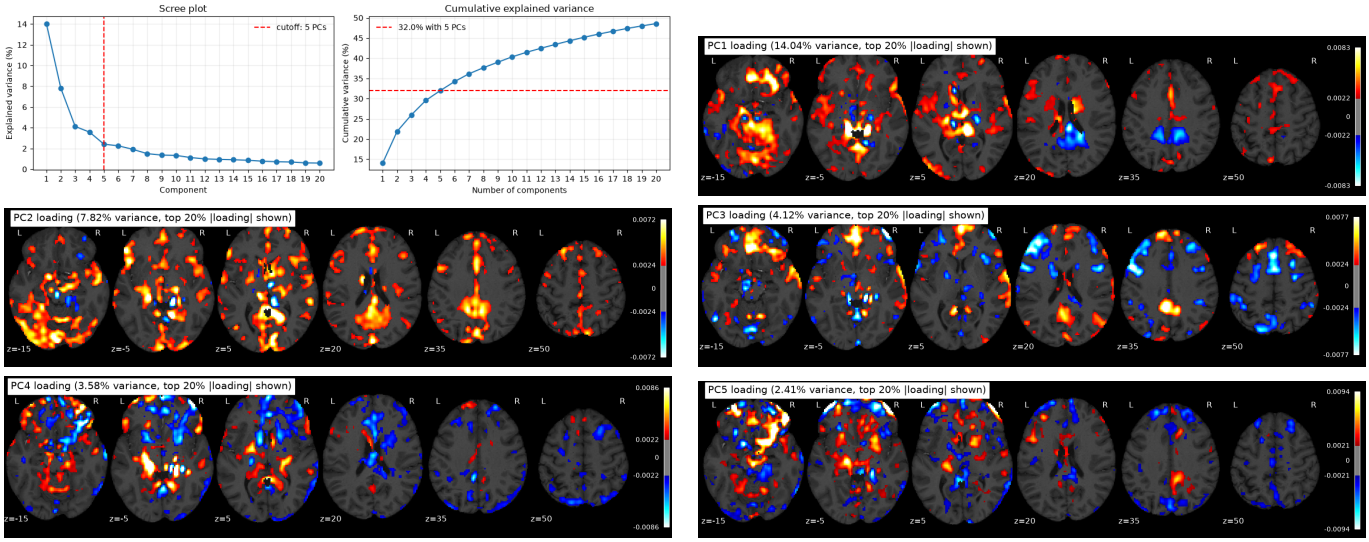


Figure 7: PCA. Top left: explained and cumulative variance of the principal components. Other panels: axial views of the spatial loading maps of PC1–PC5.

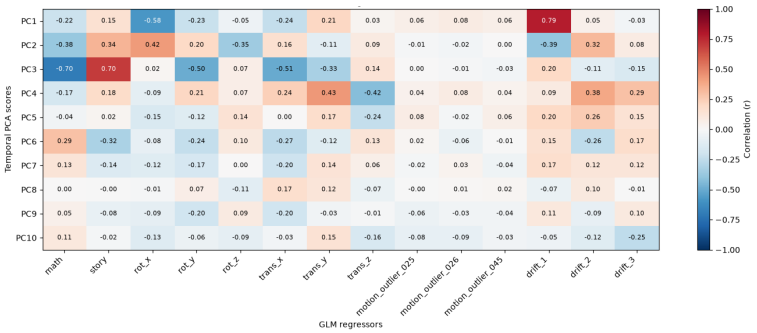


Figure 8: Pearson correlations between the first 10 PCA scores and GLM regressors.

2.2 Theoretical questions

Q1. One voxel as one sample. The data matrix becomes approximately 10^5 voxels $\times$ 316 timepoints, with at most 316 nonzero principal components (limited by the number of timepoints). Each component now represents a temporal pattern (316 loadings), while the corresponding scores describe its spatial distribution across voxels. Unlike the original approach, where components are spatial patterns and scores are time courses, the roles are reversed. The temporal components are orthogonal, and their spatial scores are uncorrelated. The PCA also differs because centering is now performed across voxels instead of time.

Q2. Comparing GLM and PCA. Comparing PCA temporal scores with GLM regressors is appropriate for exploratory interpretation, but not for statistical inference. Both methods represent fMRI data through combinations of spatial patterns and temporal signals. However, the GLM is hypothesis-driven, using predefined task regressors to identify statistically significant activations, whereas PCA is data-driven, extracting orthogonal components ordered by explained variance without distinguishing task activity from noise. Correlations with GLM regressors can suggest task-related components but do not establish statistical significance. The GLM is preferable for testing specific hypotheses, while PCA is useful for exploring dominant patterns, dimensionality reduction, and identifying potential artifacts.

Contributions. *Part 1* – structural preprocessing: Ella; functional preprocessing: Blanche; GLM: Ella; theoretical questions: Blanche, Ella. *Part 2* –PCA (variant 1): PCA code and correlation with GLM: Yassine, Zeyneb; component selection and loading maps: Zeyneb, Alae; theoretical questions: Blanche; Slides, video editing: Alae

References

Avants et al. (2008). Symmetric diffeomorphic image registration with cross-correlation. *Med. Image Anal.*; Binder et al. (2009). Where is the semantic system? A critical review and meta-analysis of 120 functional neuroimaging studies. *Cereb. Cortex*; Binder et al. (2011). Mapping anterior temporal lobe language areas with fMRI. *NeuroImage*; Corbetta & Shulman (2002). Control of goal-directed and stimulus-driven attention in the brain. *Nat. Rev. Neurosci.*; Dehaene et al. (2003). Three parietal circuits for number processing. *Cogn. Neuropsychol.*; Poldrack et al. (1999). Functional specialization for semantic and phonological processing in the left inferior prefrontal cortex. *NeuroImage*; Smith (2002). Fast robust automated brain extraction. *Hum. Brain Mapp.*; Visser et al. (2010). Semantic processing in the anterior temporal lobes: A meta-analysis of the functional neuroimaging literature. *J. Cogn. Neurosci.*; Yeo et al. (2011). The organization of the human cerebral cortex estimated by intrinsic functional connectivity. *J.*

Neurophysiol.; Zhang et al. (2001). Segmentation of brain MR images through a hidden Markov random field model and the EM algorithm. *IEEE TMI*.

AI use statement.

We used Claude (Anthropic) during the project, for both Part 1 and Part 2. It was used for coding assistance and debugging, to apply the lab code to this dataset, to plot some figures and to review code. It was also used to help interpret the results and give ideas for the theoretical questions and to improve the wording of the report. The parts most substantially influenced by AI-generated content are the code and the figures. All AI-assisted output was verified: the code was compared with the lab notebooks, simplified where needed, run end-to-end and checked. Analysis choices (BET parameters, registration method, high-pass cut-off, contrasts, thresholds, atlas, number of PCA components) were made and justified by the group based on the labs and our own results, and all text was revised by us.

Appendix

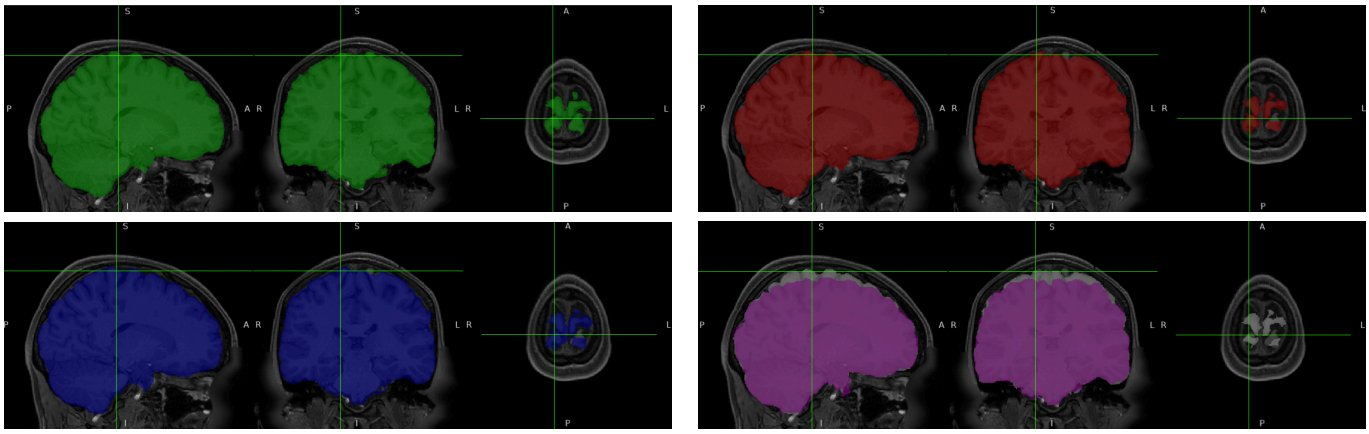


Figure 9: Comparison of BET settings on the T1w (sagittal, coronal and axial views at an example location near the top of the brain). (a) Green: robust centre estimation (`-R`) with $f = 0.5$ (selected); (b) red: default ($f = 0.5$, without `-R`); (c) blue: $f = 0.4$, without `-R`; (d) purple: vertical gradient (`-g 0.2`), $f = 0.5$, without `-R`. The vertical gradient clearly cuts the superior cortex (grey tissue above the mask in the sagittal and coronal views, almost no mask in the axial view). The default and $f = 0.4$ masks cover the superior cortex slightly less well than `-R`, with small gaps visible in the coronal and axial views.

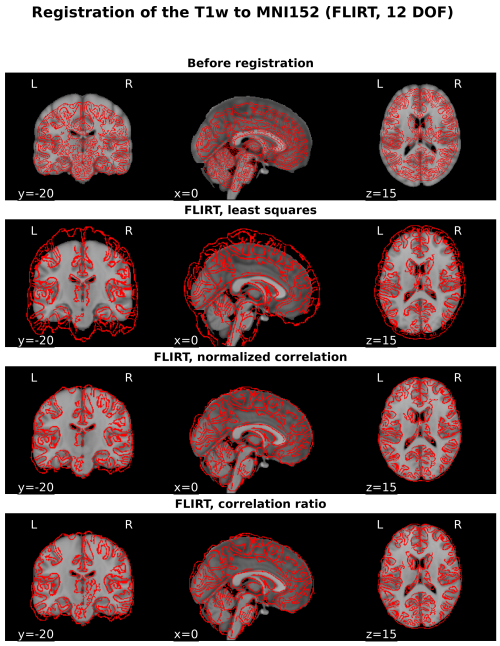


Figure 10: FLIRT (12 DOF) with three cost functions: edges of the registered T1w (red) on MNI152. Correlation with the template: least squares 0.48, normalised correlation 0.63, correlation ratio 0.62. All three affine registrations remain less accurate than the non-linear ANTs SyN registration ($r = 0.86$, Fig. **??**c).

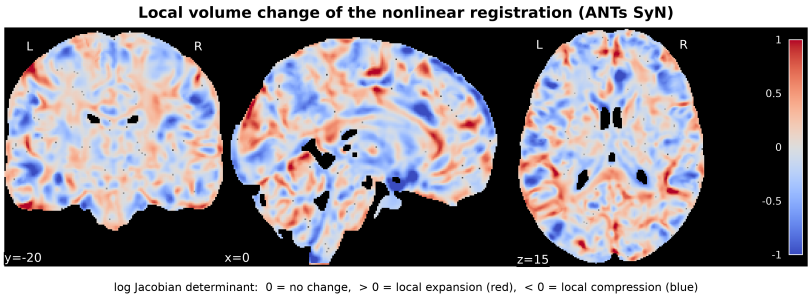


Figure 11: Log Jacobian determinant of the ANTs SyN warp (red: local expansion, blue: local compression). The Jacobian stays positive in the brain (min 0.09; 1st–99th percentile 0.35–2.31), so no folding occurs.

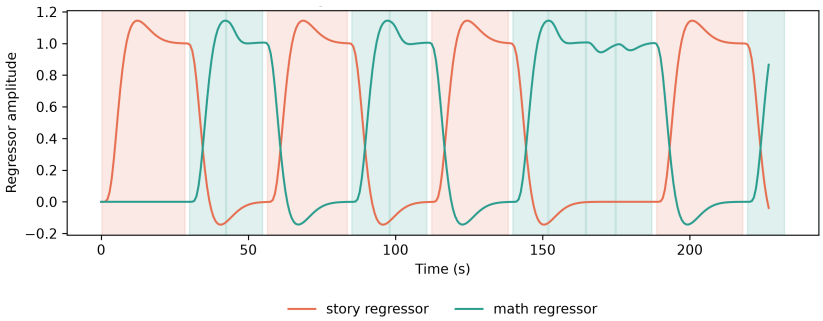


Figure 12: Story and math regressors convolved with the HRF; shaded areas mark the task blocks. The two conditions alternate with almost no rest period.